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An Investigation into Immersive Contemporary Popular Music Mixing Practice Through Analysis of Released Music

Andrew Bourbon[1], Matthew Watson[1], Dawid Stelmach[1], Michael Champion[1], Ansh Chomal, [1,] Steven Fenton[1] and Hyunkook Lee[1]

[1]University of Huddersfield

Correspondence should be addressed to Andrew Bourbon (a.j.bourbon@hud.ac.uk)

ABSTRACT

Dolby Atmos has become established as the premier format for the distribution of immersive audio since the launch becoming commercially available in June 2021. The music industry has embraced this format, with over 80% of listeners engaging with spatial audio content through a specific provider according to data released in January 2023. With this rapid development of content, we have also seen significant development in mix practice, with engineers having to develop systems and workflows for the creation of this immersive content. To better understand and document this developing practice, the authors of this paper have undertaken analysis of the 7.1.4 channel-based render of 100 commercially released Atmos mixes hosted in a Hits in Spatial Audio playlist. 2 sets of data capture and analysis have taken place. The first set of analysis has been undertaken by the authors who have listened to these 100 tracks, analysing by speaker pairs which elements are present in the mixes and confirming through a rigorous verification process. This has led to data demonstrating the positioning of elements in speakers across the collection of 100 tracks and has provided significant insight into the way in which engineers are approaching the immersive system at this point in the development of immersive music mixing practice. Subjective data has also been captured relating to musical use of the speakers and in capturing general impressions of the mix approach taken by the engineer in response to the music. The second set of analysis has taken place in MATLAB, where specific features of the 7.1.4 files have been analysed. The data generated in MATLAB includes speaker loudness, correlation between speakers, percentage of time the speaker is in use and spectral brightness. The data will be presented both as averages across the 100-track sample, and also highlighting individual tracks and tracks that conform to specific trends. Through the combination of subjective and objective analysis the authors have been able to create a detailed analysis of current mix practice, identifying established practices and demonstrating how the immersive system is being used in practice by engineers. This paper will present the data captured in this study and will discuss findings that explore both general practice across the 100 mixes and explore specific tracks in more detail to provide context as to the musical approaches embraced, linking measured practice with musical mixing and psychoacoustics. This paper is intended to be the first step in a longitudinal study that continues to measure the development of immersive popular music mixing and to inform ongoing practice through rigorous analysis of this developing format. This paper will focus on the practice discovered through the study, identifying key practices, and identifying current trends in mix approach.

1 Introduction

Spatial audio has gained significant popularity over the past couple of years, with a key contribution being the introduction of spatial audio on streaming services. The experience has become more accessible to consumers. Dolby Atmos is at the forefront of object-based formats used for spatial audio mixing and unlike previous immersive formats, Atmos has seen immense support from the industry with major studios such as Abbey Road, RAK, Metropolis to name a few investing in Dolby Atmos mix rooms with at least options for 7.1.4. monitoring, alongside offering services for immersive mixing and mastering. However, due to the novelty of spatial audio application in commercially streamed music there is limited resources and literature available documenting the practice of spatial audio mixing. This suggests an empirical study analysing spatial mixes is necessary to establish practices.

This paper aims to gain a greater understanding of how mix engineers are mixing current popular music in the Dolby Atmos in the hopes of developing a greater understanding of how the format is used in the context of music. Therefore, allowing practitioners to study spatial audio mixing rather than take a purely heuristic approach. It is also hoped that this research

2 Research Questions and Hypothesis

The following research questions and hypothesis have been established based on initial findings and observations when conducting this research project.
- How often is the Centre and LFE channel used in Dolby Atmos pop music production?
- In Dolby Atmos pop music production drums are positioned in front of the listener.
- Height channels within Dolby Atmos pop music productions contain reduced low frequency content than surround channels.
- Vocals within Dolby Atmos pop music production are less speaker locked than other elements.
- Front Height and Rear Height channels are highly correlated in Dolby Atmos pop music productions.
- The Left Right speaker pair is the most used pair of speaker within Dolby Atmos pop music productions.

3 Review of Literature

Interviews with mixing engineers who are now working within Dolby Atmos have uncover a debate concerning practice and workflow. Reliance on the stereo mix has been identified as a consistent element of creating Atmos mixes. An example of this are the workflows presented by Chris Lord Alge and Bob Clearmountain who first create the stereo mix and then enhance for Atmos. Lord Alge does this by creating stems through his hardware mix setup to then be positioned within the Atmos environment [1]. Clearmountain's approach is to complete the stereo on long fader of his console and the short fader is a post fader copy of the mix which is routed via a bus matrix to designated speakers, that allows for full 360 positioning using a channel based solution [2].
Clearmountain is very wary when using the LFE send due to not having any control over how this is rendered for binaural playback, moreover, Jessy Ray Ernster states "I don't even use it because of that reason" [3]. Ernster goes on to discuss how the centre channel never gets used in his productions due to filtering that occurs from the renderer when an object is placed where centre speaker is located. On the other hand, Clearmountain states he 'loves' the use of the centre speaker particularly for lead vocals; however, it must be noted Clearmountain utilises the centre channel through a bed. Both Clearmountain and Ernster agree that reverbs, synth pads and ambience are ideal for putting into the height channels. Again, Kybert bases his Atmos mixes off the stereo working from stems made by the stereo engineer instead of the raw multitrack, in an approach similar to Lord Alge and Clearmountain. In contrast to Clearmountain's workflow, Kybert like Ernest is using a fully object-based workflow due to his belief that "the Atmos experience [is] stronger when you [use] objects as opposed to beds" [4]. A comment Wells made regarding Atmos mixes is that many of them appear to be distracting with objects moving all over the place and he would quite often keep things static but placed within the immersive field saying, "I think using it would be a mistake most of the time" [5]. This shows that there is still plenty of debate between current practitioners in how to best use this immersive format, however the commonality between all is the reliance on the stereo mix to use as reference, suggesting that despite the use of an immersive sound field the stereo field is still the dominant component.

4 Methodology

Atmos mixes were ripped from Apple Music utilising 'Blackhole 16 Channel' virtual driver to route the individual speaker outputs into a DAW for recording as outlined in Figure 1.

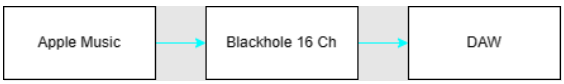

Figure 1. Capture Flow Diagram.

It was ensured when capturing mixes there was zero volume attenuation at any point in the signal chain and that Apple Music's 'Soundcheck' had been disabled to ensure no playback volume normalisation occurring. 100 tracks have been ripped using this method from the 'Hits in Spatial Audio' playlist curated by Apple Music.

This research focuses on identifying where mix engineers are placing elements within the immersive system. This led to an analysis method being developed within Microsoft Excel, that allowed for the listener to record which elements are present in different speaker pairs of the 7.1.4 system. For greater analysis on both a micro and macro level each element was then classified eg. Drums, Kick, as can be seen in Figure 2.

	Element	Classification
LEFT/RIGHT	Vocal_Lead	Lead
	Bass	Electric
	Drums	Kick
	Drums	Snare
	Synth	Pad
	Vocal_Lead	Double
	Drums	Toms
	Drums	Hats
	Reverb	Vocal Lead

Figure 2. Element and Classification Method.

This research was conducted by a group of researchers therefore to avoid error in subjective analysis a verification process was developed to mitigate this issue. This was comprised of calibration sessions between the researchers; the use of a glossary of terms to refer to; and a second listener to ensure correlation between each primary listener.

A set of objective audio features have been selected to analyse individual channels and compare correlation between different channels in Dolby Atmos mixes. The purpose of objective analysis is to support the findings of subjective analysis with potential findings of similarities between channels in respect of their activity, loudness, and spectral content.
The first measure employed was channel correlation based on rms energy which compares the similarity between channels based on channel activity. Correlation coefficient of two variables is a measure of their linear dependence.
The second measure employed was the correlation between each channel based on integrated loudness. Integrated loudness was considered most appropriate for this study as it gives an indication of the overall loudness. Loudness was calculated in accordance with ITU-R BS.1770-4 standard, channel weightings have been applied, additionally, alternative filters were used on channels such as the LFE to account for different spectral content of the channel which could otherwise skew the results of the measurement.
The last measure employed was correlation between channels based on channel brightness. In this paper the brightness of a channel is defined by the spectral centroid of a channel.

5 Subjective Results

Percentage usage of channels analysed can be seen below in Figure 3.

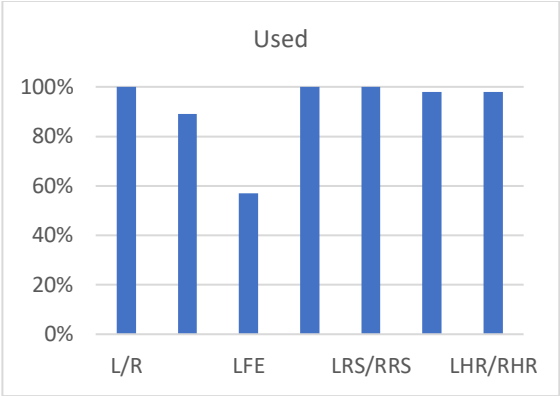

Figure 3. Percentage Use of Channels.

Speakers situated on the listener plane all see 100% usage across the sample except for the centre and LFE channels, with the centre getting used 89% of the time and the LFE only 57%. Centre channel usage may be lowered due to issues discussed by Ernster when utilising an object-based workflow. Centre channel usage identified through subjective listening also saw the use of the centre speaker as an accent speaker to reinforce specific elements found in the Left/Right pair. The low LFE usage could be partly due to the fact it is currently not rendered when using the Dolby Binaural rendering process, this is further supported by the literature as discussed in Section 3. The role of the LFE as an effect also means often generated by a large speaker with potentially slower transient response also means that the LFE is not always appropriate for use in all genres. Further results

demonstrating specific LFE usage for drums can be seen in figure 5. It has been observed that 2 tracks analysed did not utilise any of the height channels, leading to just a single plane of surround sound. An extended study would be required to verify if this is a trend or merely an outlier in the sample data. Additionally in Figure 4 below it can be seen that The Left Right pair features the most identified elements out of all the speaker pairs.

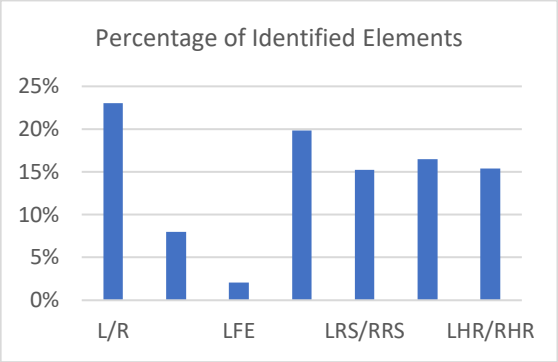

Figure 4. Percentage of Identified Elements.

Figure 4 also shows that the LFE channel has the lowest percentage of elements present. As well as being the least used of the speakers. It is interesting to note that rear surround and rear heights feature a similar number of elements identified, suggesting that there may be a link in the usage of these two speaker pairs.

Drum elements such as kick, snare and hats are heavily used in the Left-Right speaker pair being present over 50% of the time and increasing to over 70% when excluding the hats. This is also corroborated by Figure 6 which shows a detailed breakdown of the positioning of drum elements by speaker. Figure 5 clearly shows the transient underpinning of tracks, often through kick and snare are predominantly located in the L/R pair, with Kick also reinforced through the LFE in 37 of the 80 tracks where kick is identified as present. 31 of the 80 kicks are reinforced in the centre speaker and through subjective listening were more associated with the transient component of the kick drum. Of the 72 snare drums identified in the sample the snare is reinforced 19 times in the in the centre speaker but was more prevalent in the sides and rear speakers compared with the kick drum.

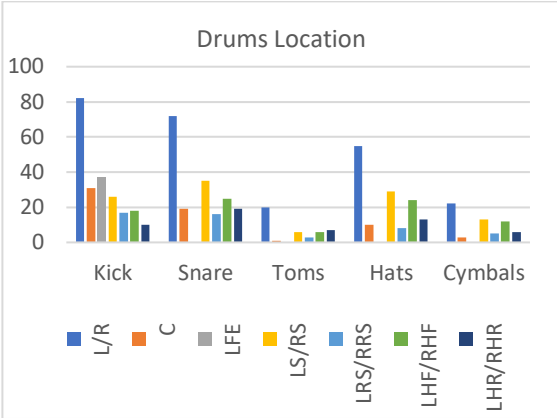

Figure 5. Drum Locations.

Figure 8 also shows that Bass of any nature is heavily placed in the Left-Right speaker pair with 88% of the tracks analysed doing so. We can also see from Figure 8 that the most prominent feature placed into the Left-Right speaker pair is the lead vocal. This suggests that despite engineers having access to a centre speaker phantom centre remains the preferred method of working when vocals are concerned.

Elements most often found in the centre channel are the kick, vocal lead, lead vocal reverb, and snare. It is interesting to note that the usage of the centre channel is low in general, with the most used element: the kick, only being present 31% of the time. Furthermore, data analysis shows the centre channel is never the most common place for an element to be placed. This suggests as discussed engineers are using the centre channel as a tool to accent the mix. Alternatively, this could be a side effect of size control when utilising objects, where objects may bleed into adjacent speakers to appear ‘larger’.

Results show the LFE gets little use within Atmos mixes, as could be expected based on its purpose of extending low frequency response. We can see that the LFE is mostly used for kick and bass elements with the former having 37% usage and the latter 41% usage. It is interesting to note that bass placed in the LFE is predominantly synthesized bass making up 31% as demonstrated in figure 6. The LFE is occasionally used for other elements however, this is in such low quantities that it can be considered insignificant.

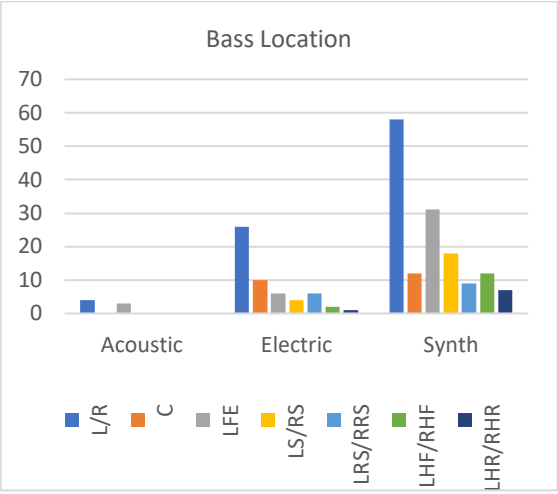

Figure 6. Bass Locations.

For the purposes of this discussion surround channels will be considered as the side and rear surrounds. The most common element found in both surround channel pairs is vocal lead reverb with 60% for sides and 56% for rears, which is significantly higher than any other element. It can also be seen that reverb in general is very high in both surround channels which is further illustrated in Figure 7.

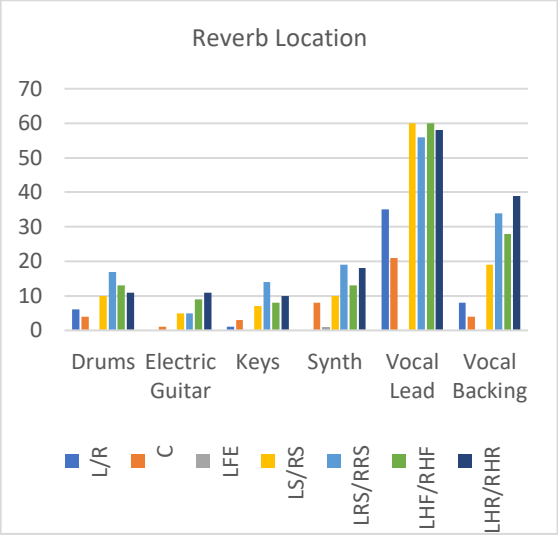

Figure 7. Reverb Locations.

Another element highly present in both sets of surround channels is vocal backing (categories: harmony and stack) with sides featuring backing harmonies 41% of the time and rears 26%. An interesting use of the sides for guitars and keys can be observed. These are elements which traditionally would be found hard-panned in the Left-Right within stereo productions and it appears engineers are taking this a step further within Atmos productions by positioning them in the side channels as can be seen in Figures 8 and 9. The keyboard distribution in figure 8 sees variance in position based on musical function when undertaking subjective listening.

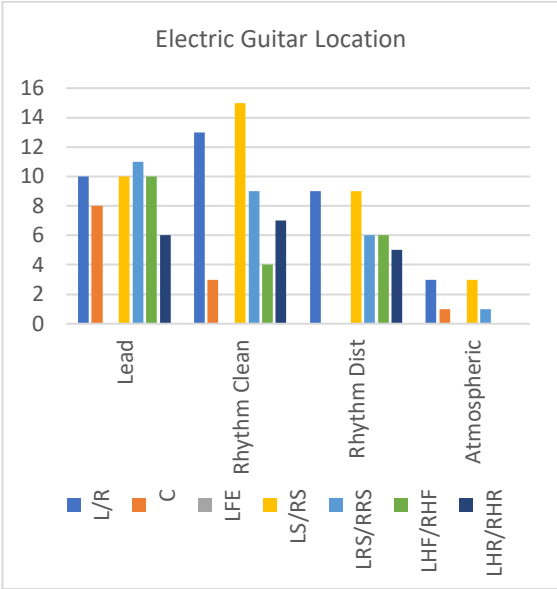

Figure 8. Electric Guitar Location.

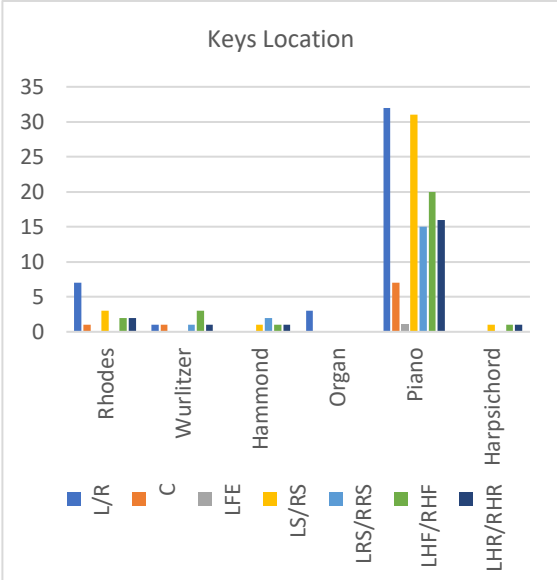

Figure 9. Keys Location.

With the height speakers, one immediate observation is the use of the front heights and rear heights is very similar. It is important to note that the Atmos Renderer will mirror the height channels when using a 7.1.2 reverb bed, which may contribute to some of the observed similarity here which will see further analysis in the objective results. The most notable difference is the placement of drums which are

mostly present in the front heights along with the lead vocal. This could possibly be an affordance of the size control parameter and further supported by frequent identification of these elements in the Left-Right speaker pair. Another key observation is the high amount of reverb present in both height channel pairs as can also be seen in Figure 7, this suggests that Clearmountain and Ernster are not the only engineers who have come to this conclusion.

6 Objective Results

In this section the objective measures performed in MATLAB as outlined in the methodology will be discussed. Table 1 below provides a key to the following correlation plots.

1	**2**	**3**	**4**	**5**	**6**
L	R	C	LFE	LSs	RSs
7	**8**	**9**	**10**	**11**	**12**
LRs	RRs	LFh	RFh	LRh	RRh

Table 1. Correlation Plot Key.

Average measures

Figure 10 shows very high correlation of rms energy between speakers within a speaker pair with all values being above 0.9.

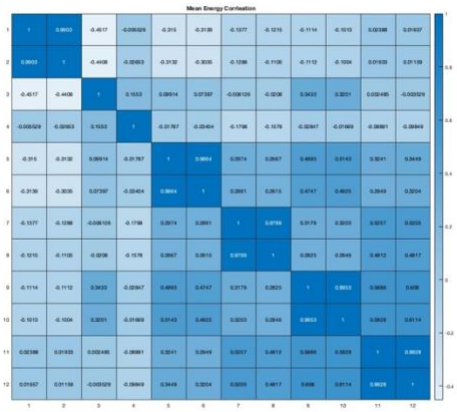

Figure 10- Mean Energy Correlation.

This indicates speakers within a speaker pair are incredibly similar. It can also be seen that there is a relatively high correlation between the front height and rear height speakers with correlation values ranging from 0.5 to 0.6. This suggests that there is a level of similarity between the front and rear height channels. Additionally, there is a medium amount of correlation between front height channels and side surround channels with correlation values around 0.5. This is similar to the correlation seen between the side surround and front height channels with correlation values again around 0.5. This could suggest that there is a similar relationship between the front heights and side surrounds as the rear heights and rear surrounds. Figure 18 also shows that there is very little correlation between the LFE and any other channel. This is to be expected due to the nature of the LFE channel. The centre channel also shows very little correlation to any other channel, which could indicate that it is being used as an isolated speaker for specific roles similarly to how the LFE is utilised as previously discussed in the objective results. One interesting observation from Figure 18 is the lack of correlation of the Left-Right with any other channel. This could suggest that the Left-Right speaker pair is effectively considered a single system with the surround and height from the other speakers in the 7.1.4 arrangement working as a second immersive system.

Figure 11 shows high loudness correlation between speakers within a speaker pair further supporting the notion of their similarity.

Figure 11- Mean Loudness Correlation.

Furthermore, a medium correlation can be observed between height channels with correlation values ranging between 0.6 and 0.7 making loudness of height channels the most correlated objective measurement in this study. No signs of correlation can be seen between any other channels.

The Left-Right speaker pair is found the loudest as shown in Figure 12, moreover, a small CI shows the Left-Right speaker pair is consistently mixed at

around –22 LUFS suggesting many mixes are still primarily driven by the Left-Right.

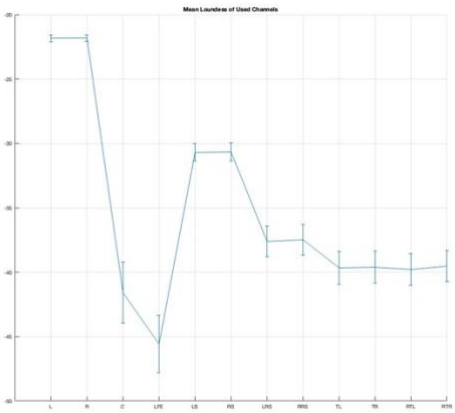

Figure 12- Mean Loudness of Used Channels.

It can be observed in Figure 12 speaker loudness decreases as we move back in the sound field. The height channels are the quietest channels (when excluding the centre and LFE) averaging at –38 LUFS. This further supports discussions suggesting the heights are an ideal location for ambience placement. The centre and LFE channels are noticeably quieter when compared to other channels. This suggests the channels have not been used consistently across mixes analysed. This is further supported by a high confidence interval (CI) (which indicates a lack of consistency) for both channels and a drastic loudness change when calculating the mean loudness for the used channels compared to all channels.

Figure 13 shows the brightness correlation between different channels, similarly to previous findings in this study a very high correlation between speakers in a speaker pair can be observed with correlation values exceeding 0.9.

Moreover, a very high correlation can be seen between the height channels with correlation values averaging 0.9 and a medium correlation between the height and side channels can be found with correlation values ranging from 0.4 to 0.5. On the other hand, the centre channel has low correlation with all the other channels with correlation values ranging from 0.1 to 0.4 and the LFE as expected has almost no correlation to any other channel with correlation values ranging from –0.2 to 0.2.

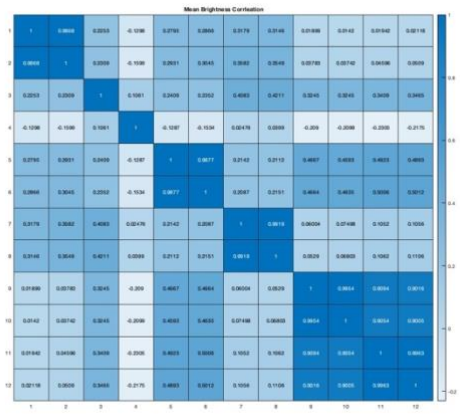

Figure 13- Brightness Correlation.

As can be seen in Figure 14 channel brightness increases as we move back in the sound field.

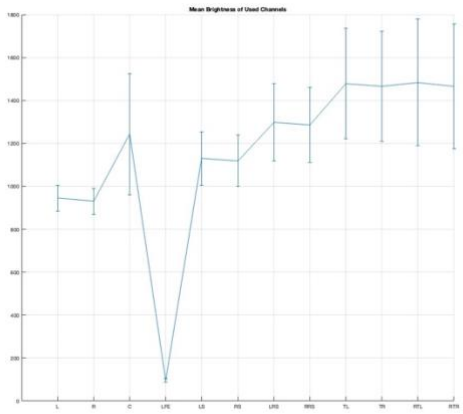

Figure 14- Mean Brightness of Used Channels.

The height channels have the highest brightness averaging at 1.5kHz. This supports the claims of placing ambient and reverberant sounds in the height channels as discussed in Section 5.1.2. However, the high CI must also be noted suggesting the use of height channels may be more versatile. The brightness of the centre channel is found surprisingly high exceeding 1.2kHz. This is most likely due to the placement of vocals and reverbs in the centre channel and reinforcement of transient. The high CI of centre channel brightness must also be noted implying a versatile use of the channel as well as a lack of consistency. As expected, the LFE channel is the darkest channel due to it only being used for the enhancement of low frequency content.

7 Analysis and Discussion

Results presented suggest that many Atmos mixes are still primarily driven by the Left-Right speaker pair. This is supported by regular identification of lead vocals, bass, and drums in the Left-Right speaker pair. Furthermore, measurements have shown the Left-Right speaker dominates Atmos mixes as the loudest pair of channels and as further can be drawn out from the data presented this is consistent throughout sample data analysed.

Although 'lead vocal' and 'kick' were elements most often located in the Centre speaker low loudness values suggest the speaker is not prominently used in Atmos mixes. This is found in agreement observations made through critical listening where the Centre speaker has often been used to accent elements in a mix.

The sparse use of the LFE in Atmos mixes may be influenced by workflow challenges as discussed previously, suggesting that engineers are concerned regarding translation between speaker and headphone-based playback. The identification of 'synth bass' in most use cases also suggests that the channel is best suited for use in specific genres.

The identification of reverb, backing vocals, and guitars in the surround speakers suggests space is used to widen the stereo image within an Atmos mix. Lower loudness values and higher brightness values as well as higher CI in the rear channels (when compared to the side channels) suggest the rears are used more for envelopment purposes whilst the sides contain more rhythmic elements with clearly defined transients. This is further supported by low correlation between speaker pairs across all three objective measures. It is also clear that height is replacing depth compared with traditional stereo staging, with engineers situating reverb significantly above the listener compared with the frontal listening plane.

Subjective analysis has uncovered frequent utilisation of height channels for reverb suggesting their primary use for providing envelopment and ambience. This is further supported by low loudness values and high brightness values. A striking similarity between front and rear height channels has been observed and is further supported by correlation values.

It is also interesting to note that there is very little movement of elements in speakers. Currently engineers are taking the stereo mix and undertaking an element driven distribution in space rather than a narrative focussed distribution. With the agency for the creation of the music in this sample lying in a post-production environment, and with stereo providing the base from which the mix is completed this is to be expected until engineers have more freedom for narrative exploration or composers and producers natively produce content in an immersive format.

8 Conclusions

To conclude, this paper outlines a start to documenting Dolby Atmos mix practices by proposing a methodology and analysis framework which can be continued in future research in both specific genre studies and more detailed track analysis. This paper has also found novel results for the usages of speakers within 7.1.4 Dolby Atmos mixes, which has answered all research questions and hypotheses laid out and can be used going forward to both educate and inform industry. The results of this paper may lead into future tools being developed for both mixing and mastering inside of Dolby Atmos.

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